

The stated numerical ratio in Conjecture 996 is false

Local AI-assisted solution draft
3 October 2026

Source. Conjecture 00000000996 in The Last Math Competition was read at the source commit

`efab34b80a63963991d6c7ed625442a89a328a44`.

The statement explicitly specifies two spectral dimensions $4/3$ and $7/4$ and claims their ratio is $7/9$.

Disproof. The asserted numerical identity fails in exact rational arithmetic:

$$\frac{4/3}{7/4} = \frac{4}{3} \frac{4}{7} = \frac{16}{21} \neq \frac{7}{9},$$

since $16 \cdot 9 = 144$ whereas $7 \cdot 21 = 147$. Reading “ratio between” in the opposite order does not help:

$$\frac{7/4}{4/3} = \frac{21}{16} \neq \frac{7}{9}.$$

Therefore the conjunction of the dimensions as written and the claimed ratio cannot hold. This disproof does not require a claim about whether the two individual spectral-dimension values are themselves correct.

Formal verification and scope. The self-contained Lean 4.19.0 file `Main.lean` uses the standard library’s exact rational type, `Std.Internal.Rat`. It defines the two dimensions and the claimed ratio directly from the numerals in the source, proves the forward and reverse ratio values, and proves `Conjecture996.conjecture996_false`. The reversed reading is also formally refuted. All reductions are kernel-checked, with no proof placeholders, custom axioms, or native decision procedures. The only reported foundational axiom is standard propositional extensionality.

This is a disproof of the statement’s explicit arithmetic equality. It establishes no additional percolation or spectral-dimension theorem. If the organizers intended a different ratio or different values, that would be a revised conjecture.

Status. This is a locally checked draft, not an organizer-accepted result. Prior submission status must be refreshed before publication.